SR UNIVERSITY

AI RESEARCH PAPER ASSISTANT

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PROBLEM STATEMENT:

- Large number of research papers published daily
- Difficult to read and understand lengthy documents
- Time-consuming manual summarization
- Hard to extract key insights and important sections
- Need for an intelligent automated solution

STAKEHOLDERS:

- Students – for easy learning and understanding
- Researchers – for quick paper analysis
- Academicians – for teaching and reference
- Data Scientists – for extracting insights
- Organizations – for research and developmen

DATASET AND MODEL DISCUSSION

- Dataset: Research papers in PDF/Text format
- Data Processing: Text extraction and preprocessing
- Models Used:NLP (Natural Language Processing)
- Transformer Models (BERT/GPT)
- RAG (Retrieval-Augmented Generation)
- Tools & Libraries: HuggingFace, LangChain, FAISS

RESULTS:

- Generated accurate summaries of research papers
- Extracted keywords and important topics
- Provided question-answering functionality
- Reduced time required for understanding papers
- Improved efficiency and productivity

APPLICATION (CLOUD HOSTED)

- Built using Streamlit web framework
- Deployed on cloud platforms (AWS / Render)
- User uploads research paper
- System processes and displays results
- Accessible anytime and anywhere

CONCLUSIONS:

- Simplifies research paper analysis
- Saves time and effort
- Enhances learning and research productivity
- Scalable and efficient system
- Future Scope:Multi-language support
- Voice-based interaction
- Advanced search features

REFERENCES:

- HuggingFace Documentation
- LangChain Documentation
- FAISS DocumentationStreamlit Documentation
- Research Paper: Lewis et al., 2020 (RAG Model)